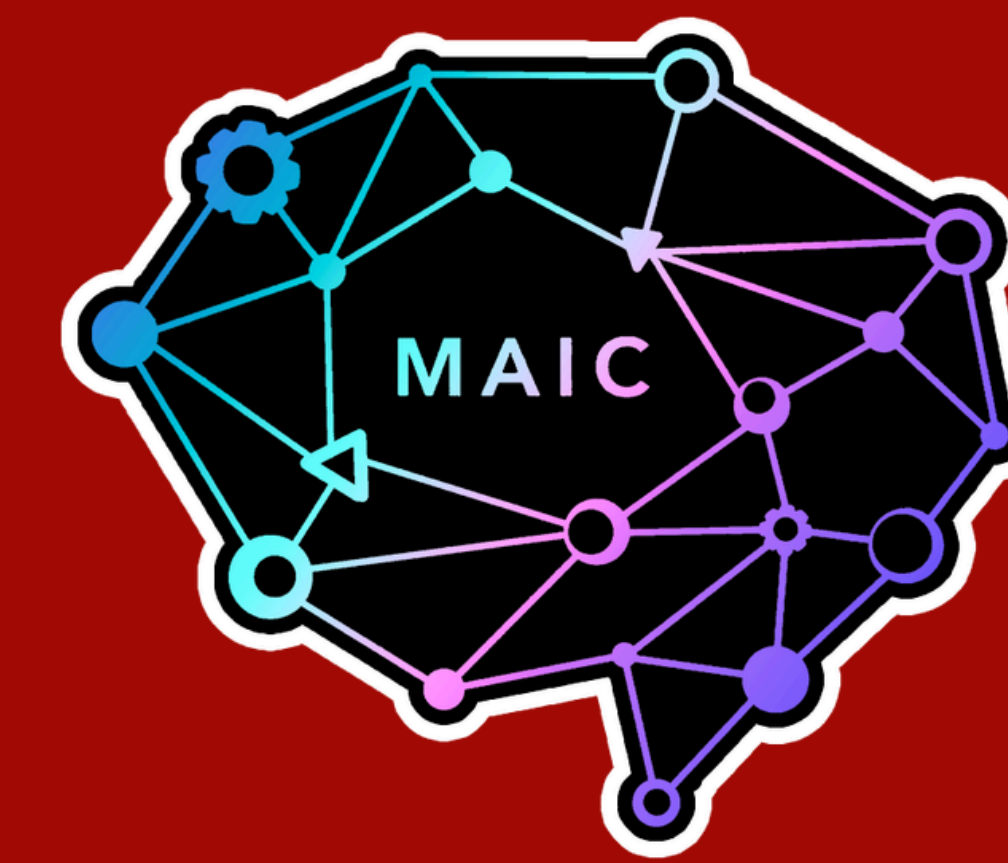


Unpacking Music Transformers: Adapting Language Model Interpretability to Musical Sequences

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Motivation



The Black Box Problem

The core problem in the interpretability space is the idea of the **black box problem**, where a complex system is trained that performs well, but we do not fully understand “why” it performs well. Interpretability aims to unpack the outputs of a model, so you can (1) explain the causes of incorrect behavior; (2) validate the internal behavior of models to reflect generalized behavior; (3) design well-informed experiments to attempt to improve model performance/reduce errors.



vs



Cross Domain Comparison

- The interpretability space is largely underexplored for music compared to language
- Insights gained from music can promote new greater progress cross-domains

Language

- Larger vocabulary (words/tokens)
- Often single correct response (generally)
- Underlying patterns are more grammatical than mathematical

Music

- Smaller vocabulary (notes/durations)
- Many correct possible responses
- Preliminary underlying mathematical patterns already known

Related Works and Foundations

Attention Is All You Need (2017): Introduces the transformer architecture based on self-attention. It served as the guiding principle behind our decoder-only implementation of the transformer architecture involving the musical domain.

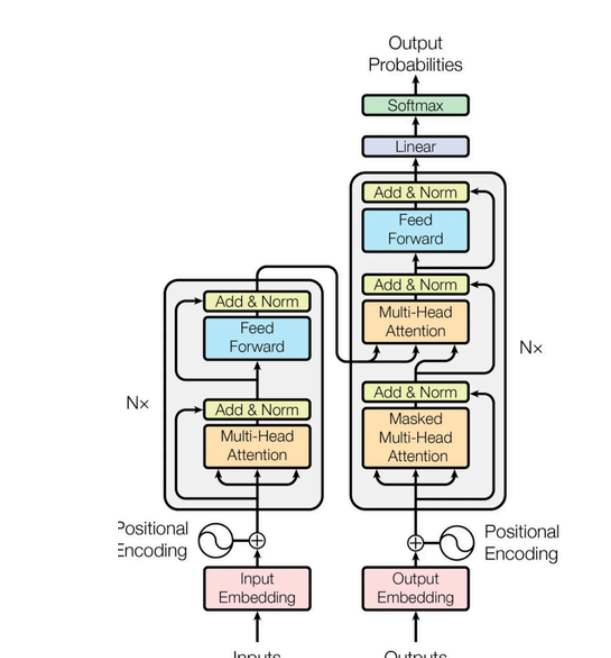
Music Transformer: Generating Music with Long-Term Structure (2018): Introduces a decoder-only transformer for MIDI music generation and the MAESTRO dataset for training.

A Mathematical Framework for Transformer Circuits (2021): Focused on how each component of a model propagates information from previous tokens in order to predict the next token. We used this to help understand how attention components influenced the prediction of tokens based on prior context.

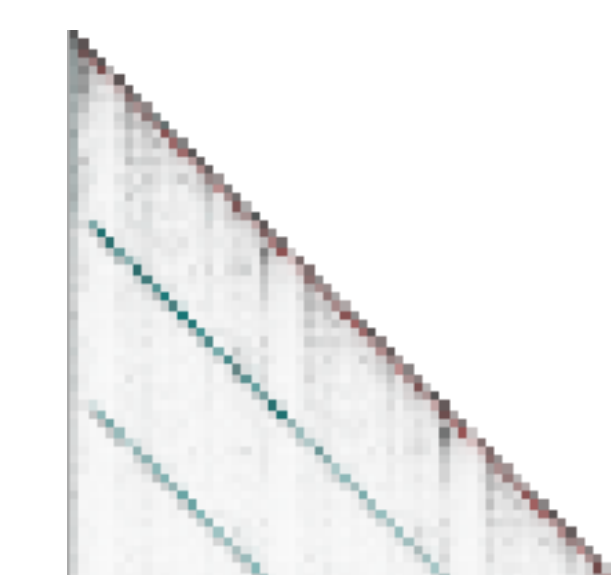
In-context Learning and Induction Heads (2022): Showcased how induction heads are able to detect patterns in the input sequence and then apply those patterns to predict the next token in a sequence. This gave us the understanding how patterns are learned in a transformer and we applied these concepts shown through natural language to our musical transformer in which patterns are prevalent throughout the entire sequence.

How to Use and Interpret Activation Patching (2024): Described the process of activation patching, similar to ablation, in which activation of a model are swapped out for cached activations from a different run. This gave us the understanding needed to perform ablation studies.

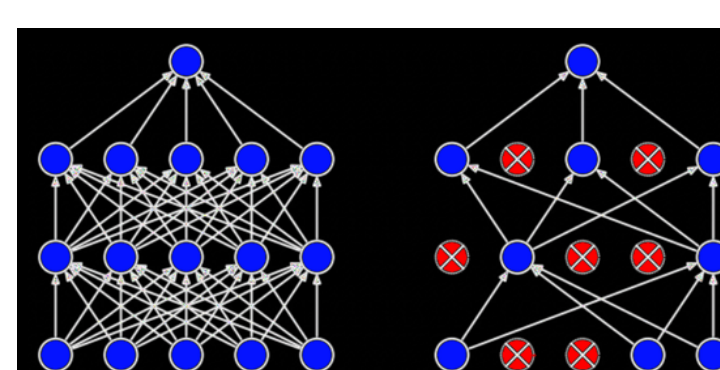
Patchscopes: A Unifying Framework for Inspecting Hidden Representations of Language Models (2024): Highlights the process of patchscoping which employs the model to create natural language explanations of their intermediate variables. This gave us the groundwork needed to perform our patchscoping study on our musical transformer to determine if the technique applies to the musical domain.



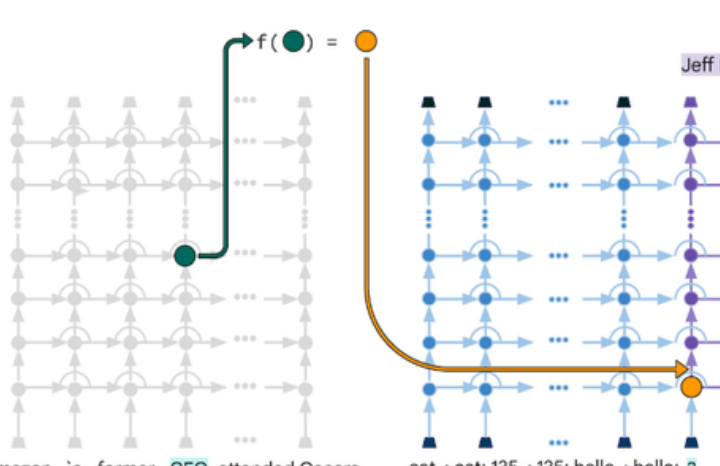
Transformer Model



Induction Head

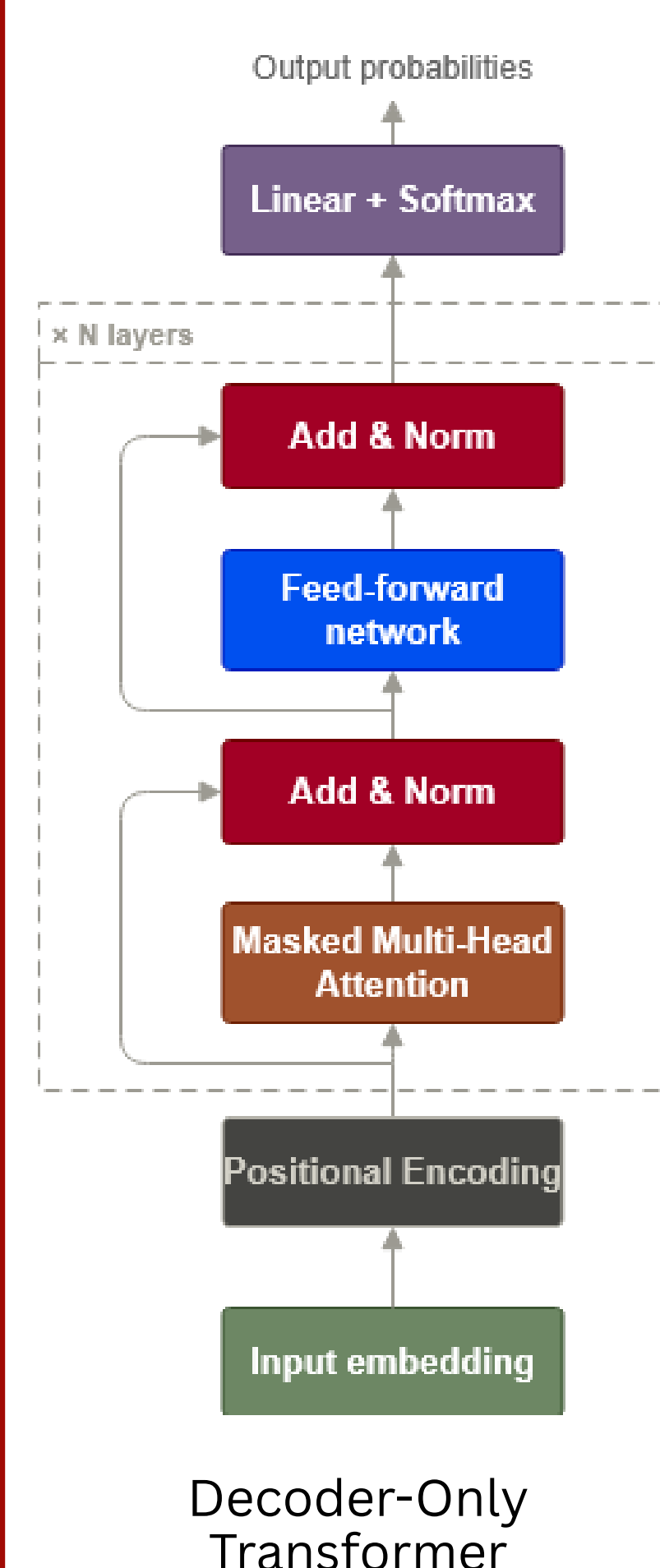


Ablation



Patchscoping

Model Architecture



The model used is a decoder-only transformer, in which each layer consists of a masked self-attention portion followed by a feed-forward network. This structure can be layered to form deeper models. In the Induction Head Detection configuration, the feed-forward portion was removed to make for easier analysis. This architecture was chosen to align with existing research on language transformers, enabling direct comparison of behaviors across the language and music domains. Key hyperparameters are listed below.

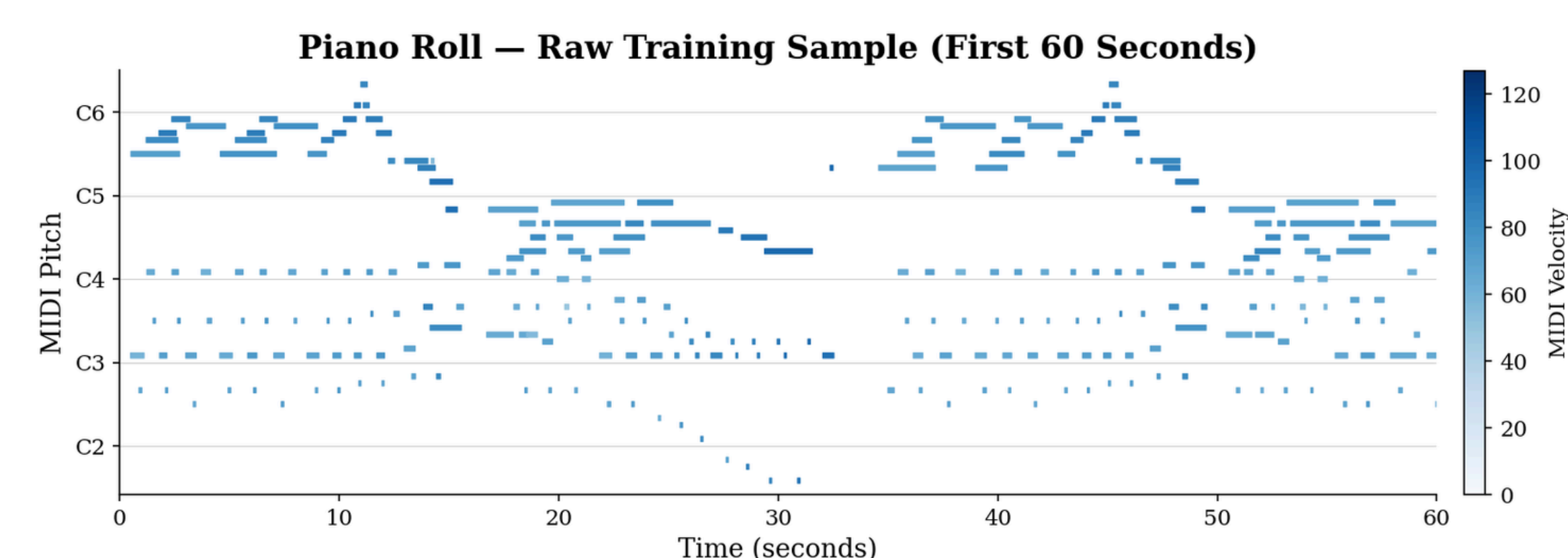
Parameter	Value	Parameter	Value
Hidden Dimension	768	Dropout	0.1
FFN Hidden Dimension	3072	Optimizer	AdamW
Attention Heads	12	Weight Decay	0.1-0.01

Data

Our Transformer models were trained with the MAESTRO Dataset (MIDI and Audio Edited for Synchronous TRacks and Organization). It contains over 200 hours of virtuosic piano performances captured with millisecond-level precision in standard MIDI format. Raw MIDI files were converted into discrete token sequences using an event-based vocabulary of 391 tokens. We encoded note onsets (NOTE_ON), note releases (NOTE_OFF), timing (TIME_SHIFT in 10ms increments), and dynamics (VELOCITY quantized into 32 bins) along with special start, end, and padding tokens. Sequences exceeding 512 tokens were randomly sliced to fit the model's context window and the dataset was split 80/20 into training and validation sets.

Token Range	Type	Count	Percent	Description
0–127	NOTE_ON	128	24.2%	Activate a pitch (MIDI 0–127)
128–255	NOTE_OFF	128	24.2%	Release a pitch
256–355	TIME_SHIFT	100	27.3%	Time step in 10ms increments (up to 1 sec)
356–387	VELOCITY	32	24.2%	Quantized dynamics (MIDI 0–127 → 32 bins)
388	SOS	1	--	Start-of-sequence
389	PAD	1	--	Padding (ignored in loss)
390	EOS	1	--	End-of-sequence

Descriptions of the 391 discrete tokens in vocabulary of MIDI sequences.



Raw MIDI piano roll of a single training sample over the first 60 seconds. Color intensity encodes MIDI velocity.

Patchscope

What is Patchscope?

Patchscope is a interpretability technique that injects a hidden state from one forward pass in the high confidence source sequence into a different forward pass from a neutral target sequence. It is common in natural language processing for LLMs, but currently lacks any published evidence of application within the music transformer domain.

Key Findings:

- Without patching: A3 probability in neutral target = 0.025 (rank 11)
- With patch from layer 11: A3 probability = 0.695. Exactly matching the source prediction
- Information crystallizes in upper layers (Figure 2)
- Generalizes across keys (Figure 1)

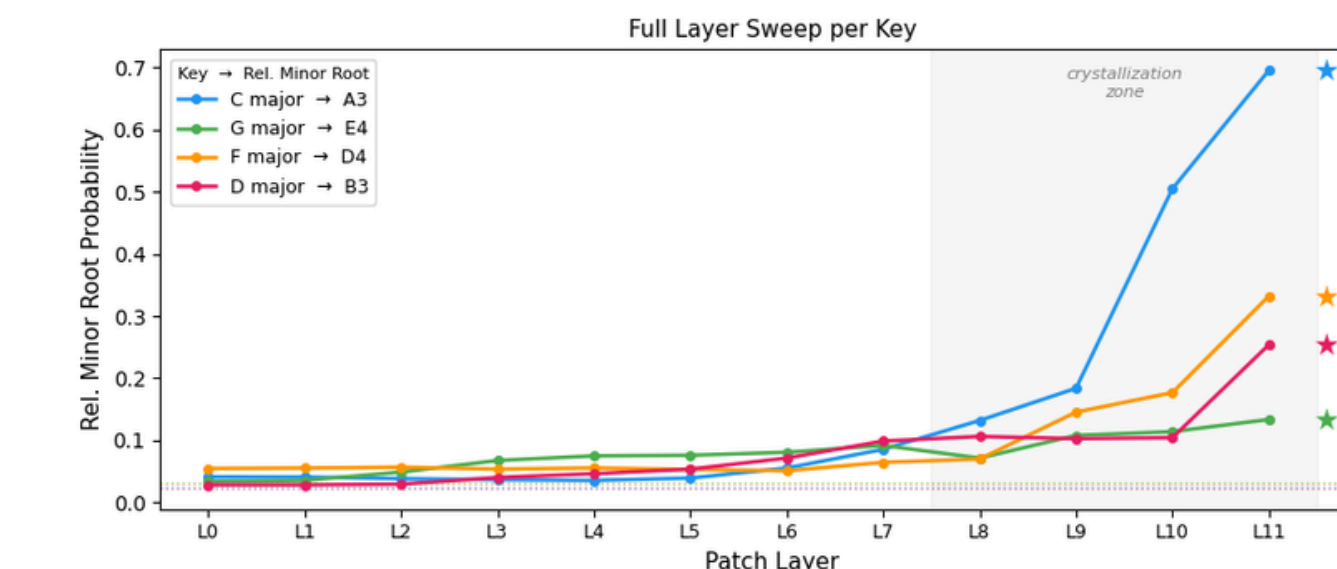


Figure 1: Patchscope layer sweep across 4 musical keys

Source context: A C-major ascending scale ending on E4. The model naturally predicts A3 as the next note (relative minor root, probability 0.695).

Target context: A neutral prompt of time-shift tokens containing no harmonic information. A3 baseline probability is only 0.025

Patch Operation: Extract the hidden state at a given layer and position from the source then inject it into the corresponding position of the target and observe whether A3 rises in probability.

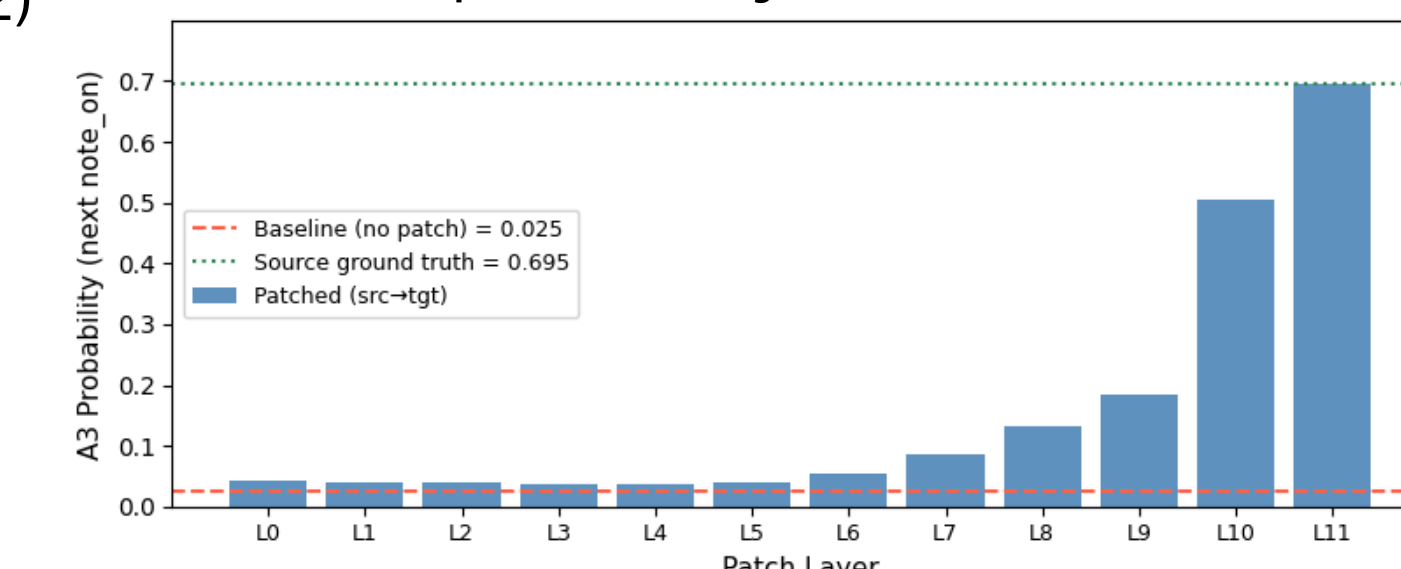


Figure 2: A3 note probability for patchscoping across all 12 layers.

Induction Head Detection

What are Induction Heads?

Induction heads are attention heads that implement a pattern of copying. Given a repeated token sequence [A][B]...[A], the head learns to predict [B] by attending back to the previous occurrence of [A].

Objective:

Do induction-like behaviors emerge in music transformers? If so, does in-context learning (ICL) also appear alongside?

Methodology:

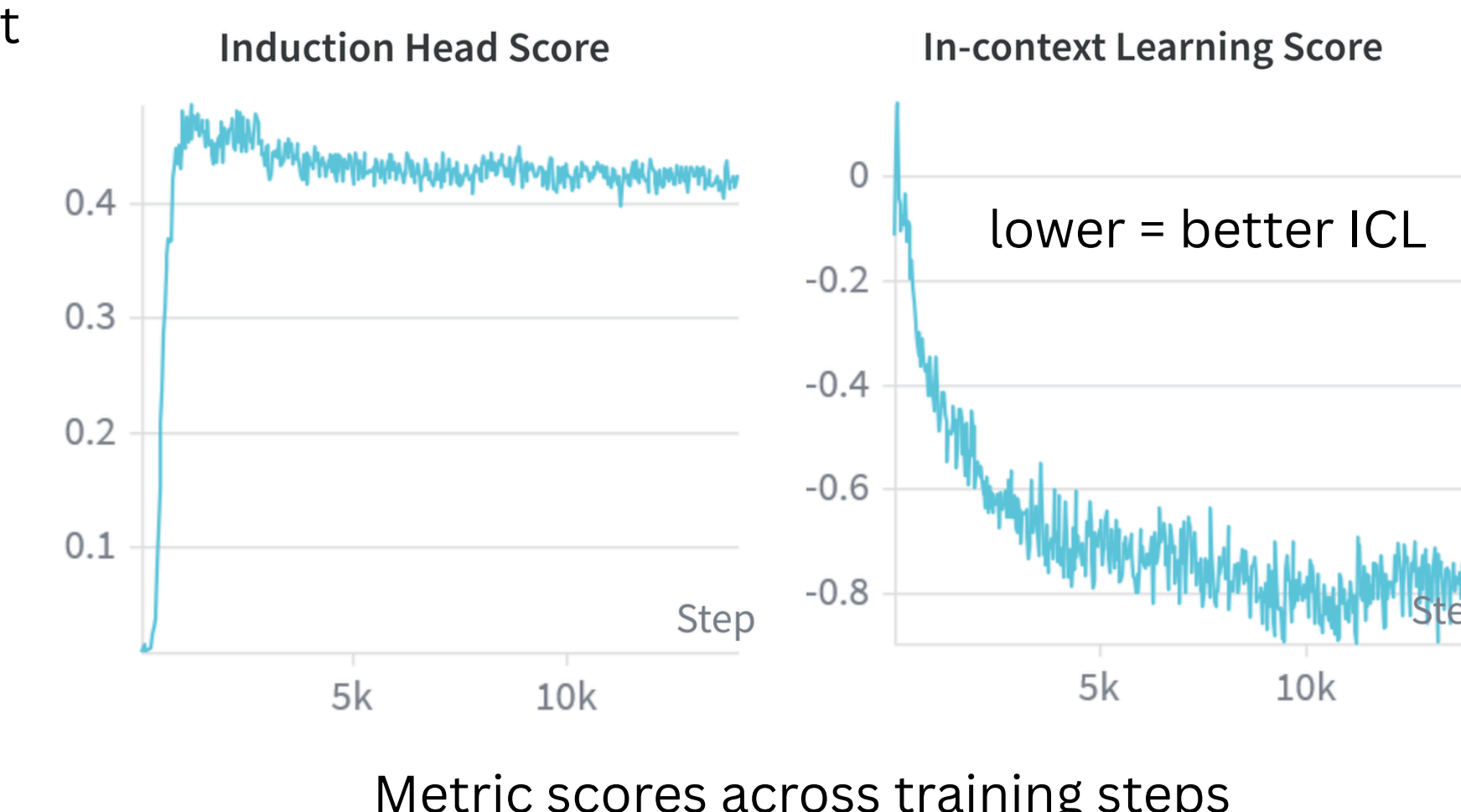
Calculate induction head score, copying logit score, previous token head score, and ICL score metrics established by Olsson et al. (2022)

Metric Results:

Induction Head Score	0.4251
Copying Logit Score	0.3228
Previous Token Head Score	0.4873
In-context Learning Score	-0.8077

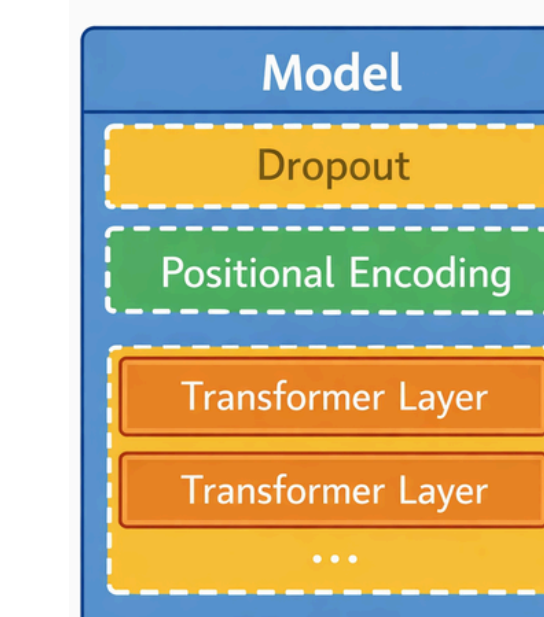
Key Findings:

Induction-like behaviors emerged in the music transformers showing similar patterns seen in language transformers. A negative ICL score provides evidence that in-context learning developed alongside induction behaviors. However, the lack of a distinct phase change across all metrics requires more research to confirm induction head formation.



Metric scores across training steps

Ablation Study



Modifiable Components from Model

Removed Component

- **Dropout (set to 0):** Without any dropout, the model had a harder time generalizing and performed worse on validation sets. The masked accuracy was overall lower than the control model.
- **Positional encoding (removed):** Without positional encoding, the model had lower masked accuracies compared to the control. It also produced sequences of music that lacked a coherent rhythm.
- **Layers in transformer (reduced):** Reducing the number of layers to 2 led to a reduced masked accuracy compared to the control model. The generated sequences of music were overall less structured and more simplistic.

Objectives:

1. Which components of the musical Transformer are critical for sequence generation?
2. How does the removal of specific components from the Transformer model impact the generated sequences of music?

Methodology:

A control model was created which retained all components of the original Transformer. Then one component was removed from the Transformer, the model was then trained and examined, and the removed component was added back.

Results:

Effect on Output